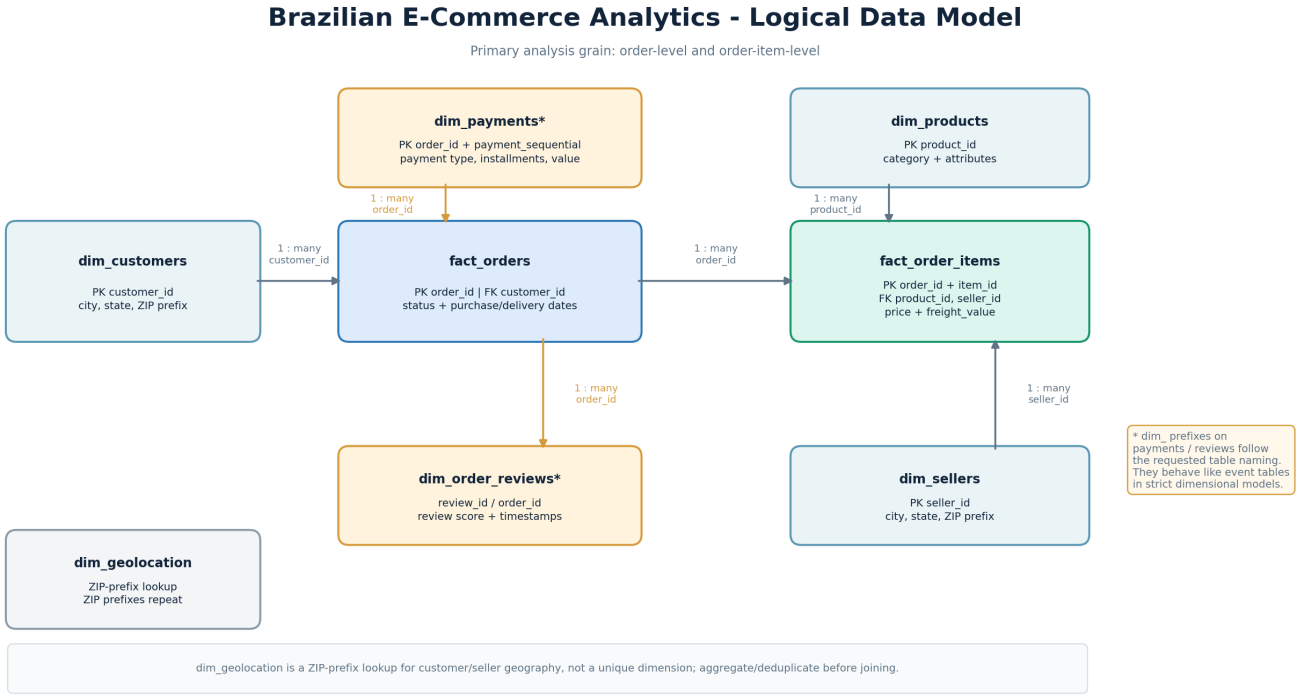
DATA ARCHITECTURE

Logical Data Model

Brazilian E-Commerce Analytics | Tables, keys, grain and join paths

The model uses the naming convention currently applied in ecommerce_db. Only fact_orders and fact_order_items have the fact_ prefix because that was the requested schema convention.



How to read the model

Arrow labels show join keys and expected one-to-many relationships. Geolocation is shown as a separate ZIP-prefix lookup, not a strict one-to-one foreign-key dimension, because ZIP prefixes repeat.

Table inventory and grain

Grain means what one row represents. Keeping each table at the right grain is essential to prevent overstated counts or monetary values.

Table	Grain	Key fields	Role / caution
fact_orders	One row per order	order_id (PK); customer_id (FK)	Order status and purchase / delivery / estimated timestamps.
fact_order_items	One row per order item	(order_id, order_item_id) (composite PK); product_id, seller_id	Item price and freight_value; can contain multiple rows per order.
dim_customers	One row per customer_id in supplied file	customer_id (PK); customer_unique_id	Customer city, state, ZIP-prefix. customer_unique_id can repeat across order-specific customer_id values.
dim_products	One row per product	product_id (PK)	Product category and product attributes. Source CSV category header contains a space: product category.
dim_sellers	One row per seller	seller_id (PK)	Seller city, state, ZIP-prefix.
dim_payments*	One row per order payment sequence	(order_id, payment_sequential) (composite PK)	Payment type, installment count and payment_value.
dim_order_reviews*	One row per review record	review_id / order_id (validate multiplicity)	Review score, comments and review timestamps.
dim_geolocation	One row per ZIP-prefix/location record	ZIP prefix is not unique	Repeated ZIP prefixes; aggregate or deduplicate before adding to customer/seller analysis.

Core relationships

Parent key	Child key	Relationship
dim_customers.customer_id	fact_orders.customer_id	One customer record to many orders
fact_orders.order_id	fact_order_items.order_id	One order to one or more item rows
dim_products.product_id	fact_order_items.product_id	One product to many item rows
dim_sellers.seller_id	fact_order_items.seller_id	One seller to many item rows
fact_orders.order_id	dim_payments.order_id	One order may have multiple payment-sequence rows
fact_orders.order_id	dim_order_reviews.order_id	An order can have review records; validate relationship cardinality

Important modelling notes

* The dim_ prefixes on payments and reviews follow the requested naming convention; semantically these are event/transaction-like tables in a strict dimensional model. Do not join raw payments and raw order_items before aggregation: one order can have multiple rows in both tables, multiplying monetary values. For state price/freight, aggregate order_items after joining through fact_orders to dim_customers. For payment_value, aggregate payment rows by the order purchase date.